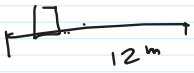
$E_{K_1} = (0.5)(150000)(6.1)$
$= 2.79 \times 10^6 J$

$E_{Kf} = (0.5)(150000)(12.8)$
$= 1.23 \times 10^7 J$

$1.23 \times 10^7 - 2.79 \times 10^6 = 9.5 \times 10^6 J$

$W = \Delta E$

$m = 49g$



3)

$W = \Delta E_K$

$E_K = \frac{1}{2}mv^2$

1. An archer pulls back her bowstring (**Figure 3**) loaded with a 22 g arrow and then releases the string. The arrow's speed as it leaves the bowstring is 220 km/h. Calculate the work done on the arrow by the bowstring. K/U [ans: 41 J]

Figure 3

$m = 22g \rightarrow 0.022 kg$ $220 km/h \rightarrow 61.11$

$V_i = 0$
$V_f = 220 km/h$
$W = ?$

$W = E_{Kf} - E_{Ki}$
$= (0.5)(0.022)(61.11)^2 -$
$= 41.078 J$

2. A space probe travels far out in the galaxy to a point where the force of gravity is very weak. The probe has a mass of 3.8×10^4 kg and an initial speed of 1.5×10^4 m/s. The probe's engines exert a force of 2.2×10^5 N in the original direction of motion as the probe travels a distance of 2.8×10^6 m. Calculate the final speed of the probe. K/U T/I [ans: 1.6×10^4 m/s]

V_f

$W = F\Delta d$

$W = \Delta E_K$

$W = E_{Kf} - E_{Ki}$

$W = (2.2 \times 10^4)(2.8 \times 10^6)$

$E_{k_i} = (0.5)(3.8 \times 10^4)(1.5 \times 10^4)^2$

$E_{k_f} = W + E_{k_i}$

$E_{k_f} = (2.2 \times 10^4)(2.8 \times 10^6) + (0.5)(3.8 \times 10^4)(1.5 \times 10^4)^2$

$(0.5)mv^2 = 4.33 \times 10^{12}$

$(0.5)(3.8 \times 10^4)\ v^2 = 4.33 \times 10^{12}$

$v = \sqrt{22824210526}$

$v_f = 1.5 \times 10^4$ m/s

3. A skater moves across the ice a distance of 12 m before a constant frictional force of 15 N causes him to stop. His initial speed is 2.2 m/s. Calculate the skater's mass. KU TI
[ans: 74 kg]

3) $v_i = 2.2$ m/s

$v_f = 0$

$\Delta d = 12$ m

$F = 15$ N

$W = 15(12)(-1)$

$\boxed{W = -180}$

$\frac{1}{2}mv_i^2 = 0 - W$

$\frac{1}{2}mv_i^2 = 180$

$(0.5)(2.2)^2 m = 180$

$m = \frac{180}{2.42}$

$m = 74.38$ kg

E1

$m = 13{,}000$ kg

$v = 3$ m/s

$E_k = (0.5)(13000)(3)^2$

$= 58{,}500$ J

$\downarrow$

$= (0.5)(15000)(3)^2$

$= 67{,}500$

Cheetah

$m = 120$ kg

$v = 33.33$ m/s

$120 \times 0.278 = 33.36$ m/s

$E_k = (0.5)(120)(33.33)^2$

$= 66{,}653$ J

2. A cat with a mass of 5.0 kg is chasing a mouse with a mass of 35 g. The mouse is running away from the cat at a constant speed in a straight line. KU TI
(a) The cat's kinetic energy is 100 times the mouse's kinetic energy. Will the cat be able to catch up with the mouse? Explain your answer.
(b) What is the minimum kinetic energy that the cat must have to keep up with the mouse?

$m_m = 0.035 kg$

$m_c = 5 kg$

$E_{kc} = 100(E_{km})$

$(0.5)(5) v_c^2 = 100(0.5(0.035) v_m^2)$

$v_c^2 = \frac{100((0.0175) v_m^2)}{(2.5)}$

$\sqrt{v_c^2} = \sqrt{0.7 v_m^2}$

$v_c = \sqrt{0.7} v_m$

$v_c = 0.84 v_m$

$v_c < v_m$

a) Cat will not be able to catch mouse

b) $E_{kc} = x E_{km}$

$(0.5)(5) v_c^2 = x(0.5(0.035) v_m^2)$

$5 v_c^2 = x(0.035) v_m^2$

$v_c^2 = \frac{x(0.035) v_m^2}{5}$

$v_c^2 = 0.007(x)(v_m^2)$

$\frac{v_c^2}{v_m^2} = 0.007(x)$

$v_c > v_m$

$\frac{v_c}{v_m} > 1$

$\frac{v_c^2}{v_m^2} > 1$

$0.007x > 1$

$x > \frac{1}{0.007}$

$x > 143.$

The cats kinetic energy has to be atleast 143 times the mouses kinetic energy.

3. A car of mass 1.5×10^3 kg is initially travelling at a speed of 11 m/s. The driver then accelerates to a speed of 25 m/s over a distance of 0.20 km. Calculate the work done on the car. K/U T/I

$m = 1.5 \times 10^3$

$v_i = 11 m/s$

$v_f = 25 m/s$

$W = ?$

$\Delta d = 0.2 km = 200 m$

$W = F\Delta d$

$= E_{kf} - E_{ki}$

$(0.5)(1.5 \times 10^3)(25)^2 - (0.5(1.5 \times 10^3)(11)^2$

$W = 378,000 J$

$W = 378{,}000\ J$

4) 4. A truck of mass 9.1×10^3 kg is travelling along a level road at an initial speed of 98 km/h and then slows to a final speed of 27 km/h. Determine the total work done on the truck. T/I

$V_i = 98\ km/h \rightarrow 27.22\ m/s$

$V_f = 27\ km/h \rightarrow 7.5\ m/s$

$W = \Delta E_K$

$= E_{Kf} - E_{Ki}$

$= (0.5)(9.1 \times 10^3)(7.5)^2 - (0.5)(9.1 \times 10^3)(27.22)^2$

$W = -3115\ 286.72$

$W = -3.1 \times 10^6\ J$

5) 5. Two objects have the same kinetic energy. One has a speed that is 2.5 times the speed of the other. Determine the ratio of their masses. K/U T/I

$E_{K_1} = E_{K_2}$

$\cancel{(0.5)}(m_1)(v_1)^2 = \cancel{(0.5)}(m_2)(v_2)^2$

$v_1 = 2.5\ v_2$

$(m_1)(2.5^2 \cancel{v_2^2}) = (m_2)(\cancel{v_2^2})$

$(m_1)(6.25) = 1\ m_2$

$m_2 = 6.25$ times m_1

$6.25\ m_1 : 1\ m_2$

6. Consider a small car of mass 1.2×10^3 kg and a large sport utility vehicle (SUV) of mass 4.1×10^3 kg. The car is travelling at 99 km/h. The car and the SUV have the same kinetic energy. Calculate the speed of the SUV. K/U T/I

$99\ km/h = 27.5\ m/s$

$E_{Kc} = E_{Ks}$

$(0.5)(1.2 \times 10^3)(27.5)^2 = (0.5)(4.1 \times 10^3)(v_s)^2$

$v_s^2 = \frac{453750}{2050}$

$v_s = \sqrt{221.341}$

$v_s = 14.87$

7) 7. An archer is able to shoot an arrow with a mass of 0.020 kg at a speed of 250 km/h. If a baseball of mass 0.14 kg is given the same kinetic energy, determine its speed. T/I

$250\ km/h = 69.44\ m/s$

$E_{KA} = \frac{mv^2}{2} = \frac{0.02(69.44)^2}{2} = 48.219\ J$

$E_{KB} = 48.219$

$48.219 = \frac{(0.14)v^2}{2}$

$\sqrt{\frac{96.438}{(0.14)}} = v$

8) 8. A hockey player shoots a puck at a speed of 150 km/h. The mass of the puck is 0.16 kg, and the player's stick is in contact with it over a distance of 0.25 m. Calculate the average force exerted on the puck by the player. T/I

$W = F\Delta d$

$v = 41.667\ m/s$

$m = 0.16\ kg$

$\Delta d = 0.25\ m$

$E_K = \frac{0.16(41.667)^2}{2} = 138.891\ J$

$W = E_K$

$$\sqrt{\frac{96.438}{(0.19)}} = v$$

$$v = 26.2 \text{ m/s}$$

$$E_k = \frac{0.16(17.667)}{2} = 138.891 \text{ J}$$

$$W = E_k$$

$$F = \frac{138.891}{0.25}$$

$$F = 555.5 \text{ N}$$

9. At room temperature, an oxygen molecule with mass 5.31×10^{-26} kg has kinetic energy of about 6.25×10^{-21} J. Determine the speed of the molecule.

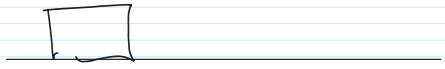
$$E_k = 6.25 \times 10^{-21}$$

$$6.25 \times 10^{-21} = \frac{5.31 \times 10^{-26}(v)^2}{2}$$

$$\sqrt{\frac{2(6.25 \times 10^{-21})}{5.31 \times 10^{-26}}} = v$$

$$v = 485.19 \text{ m/s}$$

10. A horizontal force of 15 N pulls a block of mass 3.9 kg across a level floor. The coefficient of kinetic friction between the block and the floor is $\mu_K = 0.25$. If the block begins with a speed of 0.0 m/s and is pulled for a distance of 12 m, determine the final speed of the block.



$$F_f = \mu F_N$$

$$= \mu (mg)$$

$$F_{net} = F_A - F_f$$

$$W = (F_A - F_f)\Delta d$$

$$W = E_{k2} - \cancel{E_{k1}}$$

$$W = E_{k2}$$

$$\frac{mv^2}{2} = (F_A - \mu(mg))\Delta d$$

$$\frac{3.9v^2}{2} = (15 - (0.25)(3.9 \times 9.8))\ 12.$$

$$\frac{3.9v^2}{2} = 65.39$$

$$v = \sqrt{\frac{65.39(2)}{3.9}}$$

$$v = 5.8 \text{ m/s}$$

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phys12_c04_4_2.pdf

to the Work–Energy Theorem

You ca[illegible]rk–energy theorem to solve several types of physics problems. However, you cannot control all of the variables in the real world as easily as you can in a physics experiment. The work–energy theorem is only true if no energy losses occur. In many real-world situations, energy will seem to disappear in the form of light, sound, heat, or changes in the shape of an object. For instance, in a car collision, energy goes [illegible] in the car. The work don[illegible] energy of that car.

This discussion assumes that the applied force is constant. The derivation of the work–energy theorem for a varying force requires calculus, but the result is the same. The work–energy theorem holds true, even when the applied force is not constant.

Investigation 4.2.1

The Work–Energy Theorem (page 209)

[illegible] that you hav[illegible]ned about [illegible]ork–energy [illegible]n, you are [illegible] to complet[illegible]estigation to test the theorem. This Controlled Experiment will give you an opportunity to calculate the force of friction and predict the velocity of a system.

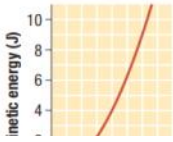
4.2 Review

Summary

- Kinetic energy, E_k, is the energy an object has due to its motion. It is a scalar quantity because no direction is associated with it. The units of kinetic energy are joules (J).
- An object's kinetic energy is related to its mass, m, and its speed, v, by the equation $E_k = \frac{1}{2}mv^2$.
- According to the work–energy theorem, the total work done on an object is equal to the change in the object's kinetic energy: $W = \Delta E_k$.

Questions

1. Could an elephant walking slowly across a field have more kinetic energy than a cheetah chasing its prey? Explain your answer. K/U T/I
2. A cat with a mass of 5.0 kg is chasing a mouse with a mass of 35 g. The mouse is running away from the cat at a constant speed in a straight line. K/U T/I
   (a) The cat's kinetic energy is 100 times the mouse's kinetic energy. Will the cat be able to catch up with the mouse? Explain your answer.
   (b) What is the minimum kinetic energy that the cat must have to keep up with the mouse?
3. A car of mass 1.5×10^3 kg is initially travelling at a speed of 11 m/s. The driver then accelerates to a speed of 25 m/s over a distance of 0.20 km. Calculate the work done on the car. K/U T/I
4. A truck of mass 9.1×10^3 kg is travelling along a level road at an initial speed of 98 km/h and then slows to a final speed of 27 km/h. Determine the total work done on the truck. T/I
5. Two objects have the same kinetic energy. One has a speed that is 2.5 times the speed of the other. Determine the ratio of their masses. K/U T/I
6. Consider a small car of mass 1.2×10^3 kg and a large sport utility vehicle (SUV) of mass 4.1×10^3 kg. The car is travelling at 99 km/h. The car and the SUV have the same kinetic energy. Calculate the speed of the SUV. K/U T/I
7. An archer is able to shoot an arrow with a mass [illegible]
9. At room temperature, an oxygen molecule with mass 5.31×10^{-26} kg has kinetic energy of about 6.25×10^{-21} J. Determine the speed of the molecule. T/I
10. A horizontal force of 15 N pulls a block of mass 3.9 kg across a level floor. The coefficient of kinetic friction between the block and the floor is $\mu_K = 0.25$. If the block begins with a speed of 0.0 m/s and is pulled for a distance of 12 m, determine the final speed of the block. T/I
11. Centripetal forces do non-zero work on objects in non-circular orbits. Satellites in non-circular orbits around Earth have different speeds at different positions in their orbit. The change in kinetic energy comes from work done on the satellite by gravity. A satellite of mass 5.55×10^3 kg has a speed of 2.81 km/s at one point in its orbit and a speed of 3.24 km/s at a second point. Calculate the work done on the satellite by gravity as the satellite moves from the first point to the second point. T/I A
12. **Figure 4** shows the kinetic energy of a robot as a function of its speed. K/U T/I



car is travelling at 99 km/h. The car and the SUV have the same kinetic energy. Calculate the speed of the SUV. K/U T/I

7. An archer is able to shoot an arrow with a mass of 0.020 kg at a speed of 250 km/h. If a baseball of mass 0.14 kg is given the same kinetic energy, determine its speed. T/I

8. A hockey player shoots a puck at a speed of 150 km/h. The mass of the puck is 0.16 kg, and the player's stick is in contact with it over a distance of 0.25 m. Calculate the average force exerted on the puck by the player. T/I

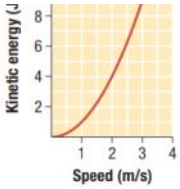


Figure 4

(a) What type of function is this?

(b) Why does the graph pass through the origin?

(c) Determine the mass of the robot.

(d) Write an equation that relates kinetic en a function of speed for the robot.